THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) south-seeking pole **(b)** neutral centre
(c) north-seeking pole **(d)** non-magnetic part

2. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) type of plant **(b)** amount of water **(c)** amount of light **(d)** amount of soil

3. Why do prey animals like deer have eyes on the sides of the head? 1

(a) to swim **(b)** wide field of view to spot predators
(c) to dig **(d)** to see only forward

4. A person eats only steamed vegetables and fruits and nothing else. The likely lack is: 1

(a) water **(b)** protein and fat **(c)** fibre **(d)** vitamins

5. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron only **(b)** wood
(c) all metals **(d)** iron and aluminium

6. Which of the following is not good practice while recording data? 1

(a) rounding off numbers to look neat without noting it
(b) writing units
(c) noting the date
(d) giving a title

7. A jungle has more kinds of organisms than a desert because: 1

(a) it is colder **(b)** it provides more food, water and shelter
(c) it has no animals **(d)** it has no plants

8. An experiment gave these results: Food X – blue-black with iodine; Food Y – oily patch; Food Z – no change in either test. Which is correct? 1

(a) X has starch, Y has fat, Z has neither starch nor fat
(b) X has fat, Y has starch, Z has both
(c) all three contain starch
(d) X has no starch, Y has starch, Z has fat

9. A student suspends magnet A and brings magnet B's north pole near A's north-seeking end. A turns away. This shows: 1

(a) A is iron
(b) B's north repels A's north
(c) B's north attracts A's north
(d) B has no poles

10. A scientist is convinced that her hypothesis is right, but the data shows otherwise. The correct action is to: 1

(a) repeat until the data matches
(b) discard the data
(c) revise the hypothesis
(d) change the readings

11. Which statement explains why camels have broad feet? 1

(a) to avoid sinking in sand
(b) to dig for water
(c) to climb
(d) to swim

12. A child gets tired quickly, has pale skin and breathlessness. Which food should the doctor suggest? 1

(a) chips
(b) butter
(c) spinach and jaggery
(d) sugar

13. **Assertion (A):** A freely suspended bar magnet always comes to rest along the north-south direction. 1
Reason (R): The Earth behaves like a huge magnet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Observation is the first step of the scientific method. 1
Reason (R): A telescope helps us observe objects that are very far away in the sky.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Fish have a streamlined body shape. 1
Reason (R): A streamlined body helps a fish to move easily through water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a dolphin look like a fish but is classified as a mammal? 2

17. Why is a diet rich only in proteins and fats not balanced? 2

18. A student says that heating a magnet strengthens it. Do you agree? 2

19. Give one example in which technology causes a problem and how science may solve it. 2

20. Why do aquatic plants such as hydrilla have thin, flexible stems? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how a loudspeaker uses a magnet. 3

22. Explain the term 'variable' and give examples of three variables in an experiment on dissolving sugar. 3

23. Explain with an example how animals show adaptation in feeding. 3

24. How does one choose healthy snacks when travelling? 3

OR

How would a magnet help sort iron nails from a mixture of iron nails and brass screws? 3

25. In a plant experiment, plants P (light, water), Q (dark, water), R (light, no water). State what each comparison tests. 3

OR

Compare the adaptations of a duck and a fish for movement in water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A class visited a garden in Jaipur and listed these plants: tulsi, rose, mango, pumpkin and grapevine. The teacher said plants can be grouped as herbs, shrubs, trees, creepers and climbers by the nature of their stems. 5

(i) Name the plant from the list that is a tree.

(ii) Which plant is a creeper?

(iii) Give one difference between a herb and a shrub.

(iv) How is a climber different from a creeper? Give an example of a climber from the list.

27. Discuss how traditional Indian meals combine foods to give a balanced diet, using thali as an example. 5

OR

Describe an activity to show that the strength of a magnet is highest at the poles, and write observations and a conclusion. 5

END OF QUESTION PAPER — Total: 50 Marks

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